



A creative and sensitive mind with strong technical foundations. Expert in Bayesian machine learning. Trying to understand the big questions. Heavy on principle. Making things with others.

REACH

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FOCUS

- Bayesian machine learning
- Theoretical foundations of inference
- Real-time sensor fusion and active inference
- Creative AI systems
- Human-machine interaction
- Interdisciplinary collaboration

TECHNICAL

- Python
- Julia
- R
- C/C++
- JavaScript
- JAX
- PyTorch
- Linux
- Docker
- Raspberry Pi
- Google Cloud

LANGUAGES

Dutch	native
English	fluent
French	intermediate
Portuguese	basic

INTERESTS

- Running
- Mindfulness
- Playing drums

BAYESIAN NONPARAMETRIC GLOTTAL INVERSE FILTERING

PhD thesis

Accurate and real-time decomposition of speech into separate contributions from the vocal folds and the vocal tract.

ON BASIS FUNCTION SELECTION FOR SPARSE GAUSSIAN PROCESS REGRESSION

Preprint

We derive an information-theoretic criterion to select basis functions for sparse GP regression.

NEXT LEVEL NAMING GAME

Pluto notebook

How do successful communication strategies emerge, and by which social networks?

A WEAKLY INFORMATIVE PRIOR FOR RESONANCE FREQUENCIES

Article

I derive a new prior for resonance frequencies from the maximum entropy principle.

GROWING NETWORKS WITH A GIVEN ASSORTATIVITY COEFFICIENT

Research notes

I propose a novel minimally biased way to sample or grow assortative networks.

SYNCRETIC ANIMACY

Essay

Are we capable of authentic interactions with animate artwork that can think and watch back?

SELECTED CREATIVE WORK

Feb 2026

SUNSTREAM.TV – Online stream forecasting the Sun in real-time

Ongoing cooperation with Jo Caimo (artist) and James Walsh (Cambridge) to predict the state of the Sun eight minutes ahead from continuously incoming satellite data. I am developing a high-quality model based on the recent SDO-FM foundation model to make this possible for the first time. I built a custom pipeline that compiles C++ code and PyTorch Lightning to WebAssembly to run the model natively in web browsers.



Apr 2025

GRAPHOMANIAC – Interactive AI that responds creatively in the moment

Online installation (click to visit) emitting a continuous stream of observations and inner thoughts about its environment. The AI has real-time vision and reacts instantly and naturally to whatever it perceives.



Nov 2024

SPELLS & PROMISES – Exposition

I conceived and led the production of this contemporary art exhibition, which brought together seven artists working with emerging technologies as artistic media. The show was picked up by several leading art blogs: ofluxo.net, fakewhale.xyz, saliva.live, and kubaparis.com.



Sep 2024

NEO SEER – 3D-printed sculpture housing autonomous AI

Collaboration with Mathias Mu (Royal Academy of Fine Arts Antwerp) where we designed a biomorphic 3D-printed sculpture hosting autonomous AI. I gathered training data, performed data augmentation, tested and deployed the LLM pipeline for continuous high-throughput processing, and installed it on an embedded Raspberry Pi with robust communication and fault tolerance. We showed this piece at several exhibitions, and it later became a subject of study by the PALETTES research group (KU Leuven).



HIGHER EDUCATION

MSC IN PHYSICS & ASTRONOMY (CUM LAUDE)

UGent

Acquired strong fundamentals in mathematical modeling and scientific coding, with thesis work published in Scientific Reports.
2010 - 2017

PHD IN COMPUTER SCIENCE

Vrije Universiteit Brussel & UAntwerpen

PhD combined with government artist grant. Working on Bayesian nonparametrics, signal processing, and network theory.
2018 - 2026 (defence in June)